

International Solar Feasibility Analysis

Abhi Andrew, Ean Brown, Kyle Corbin, Travis Moon, Madyson Stanley,
Prof. Brian Mckenna



Solar Feasibility Tool

Solar Feasibility Dashboard		Results
Project Info		
Project Name		
Location		
Roof Area (m ²)		
Electric Company		
Solar Production		
Avg Daily Solar (kWh)		
Estimated System Size (kW)		
Estimated Annual Solar Production (kWh)		
Financial Summary		
Total Installation Cost		
Annual Energy Savings		
Payback Period (years)		
Lifetime Savings		
Buy back		

The Mission: To develop a standardized, repeatable computational framework that empowers Global Strategies to independently evaluate the technical and financial viability of solar power installations across international ministry sites.

Project Rationale & Objectives

- Build a simple, repeatable tool to help Global Strategies evaluate solar for different sites
- **Scalability:** Make it scalable so it can be used across multiple locations
- **Economic Validation:** Provide clear financial “go/no-go” decisions (ROI, payback)
- **Accessibility:** Ensure long-term usability for non-technical, multi-lingual stakeholders

System Specifications

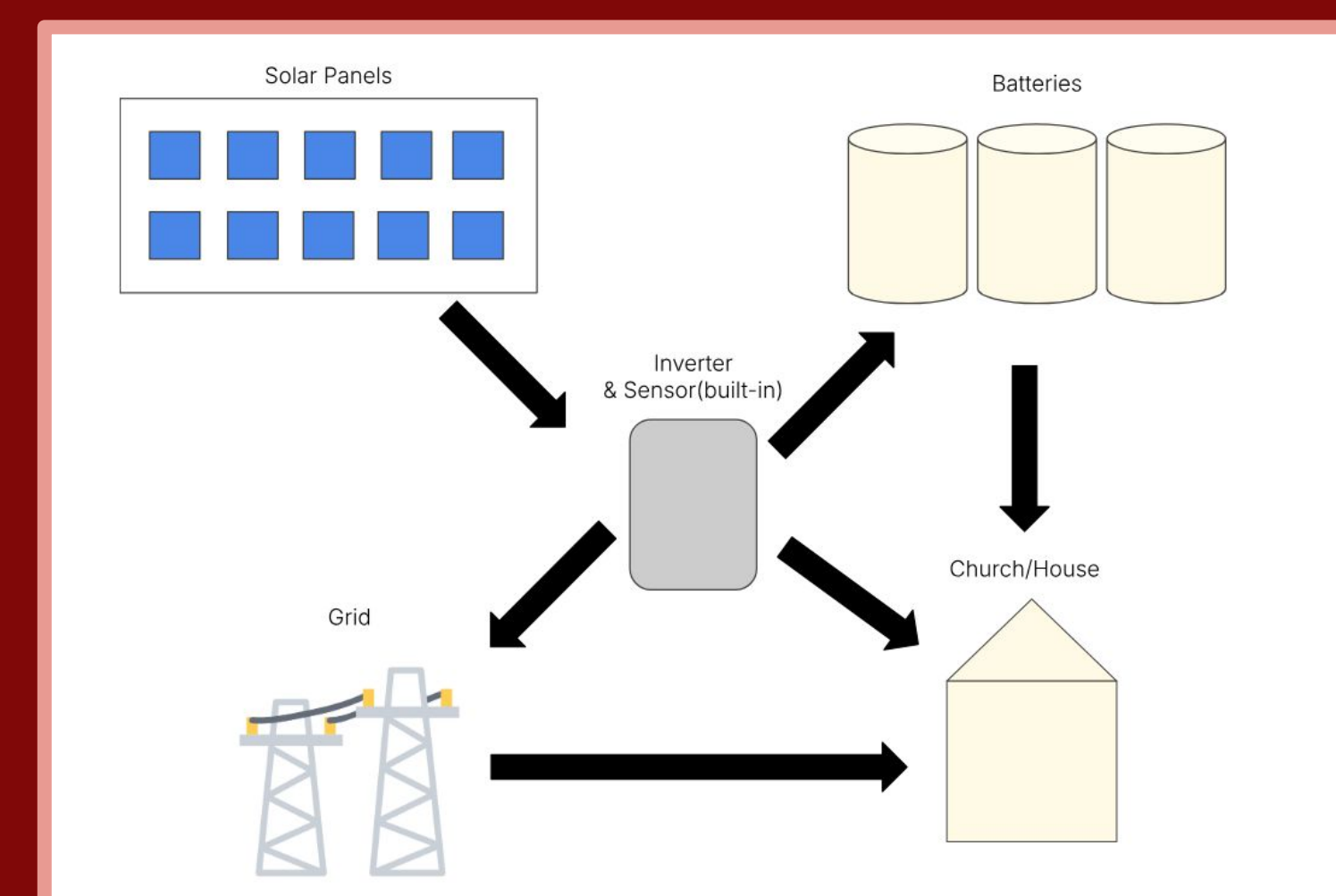
- **Platform:** Developed within Google Sheets/Excel
- **Financial Logic:** Real-time calculation of installation costs, 25-year lifetime savings, and Net Present Value (NPV).
- **Grid Integration:** Algorithmic modeling of local utility "Time-of-Use" (TOU) rates and surplus energy buyback programs.
- **Bilingual Support:** Full English and Spanish UI for local operational management

Design Methodology

- **Phase 1:** Analyze site conditions (shading, structure, energy use)
- **Phase 2:** Build the model to estimate cost, savings, and payback
- **Phase 3:** Test with real cases (Alajuelita & Torgeson) to validate results

The Solar Feasibility Tool is a computational model built to simplify energy investment decisions for international ministry sites. By processing site-specific data, such as geographic coordinates, energy consumption, and local utility rates, the tool will be able to calculate technical performance and financial ROI.

Test Device

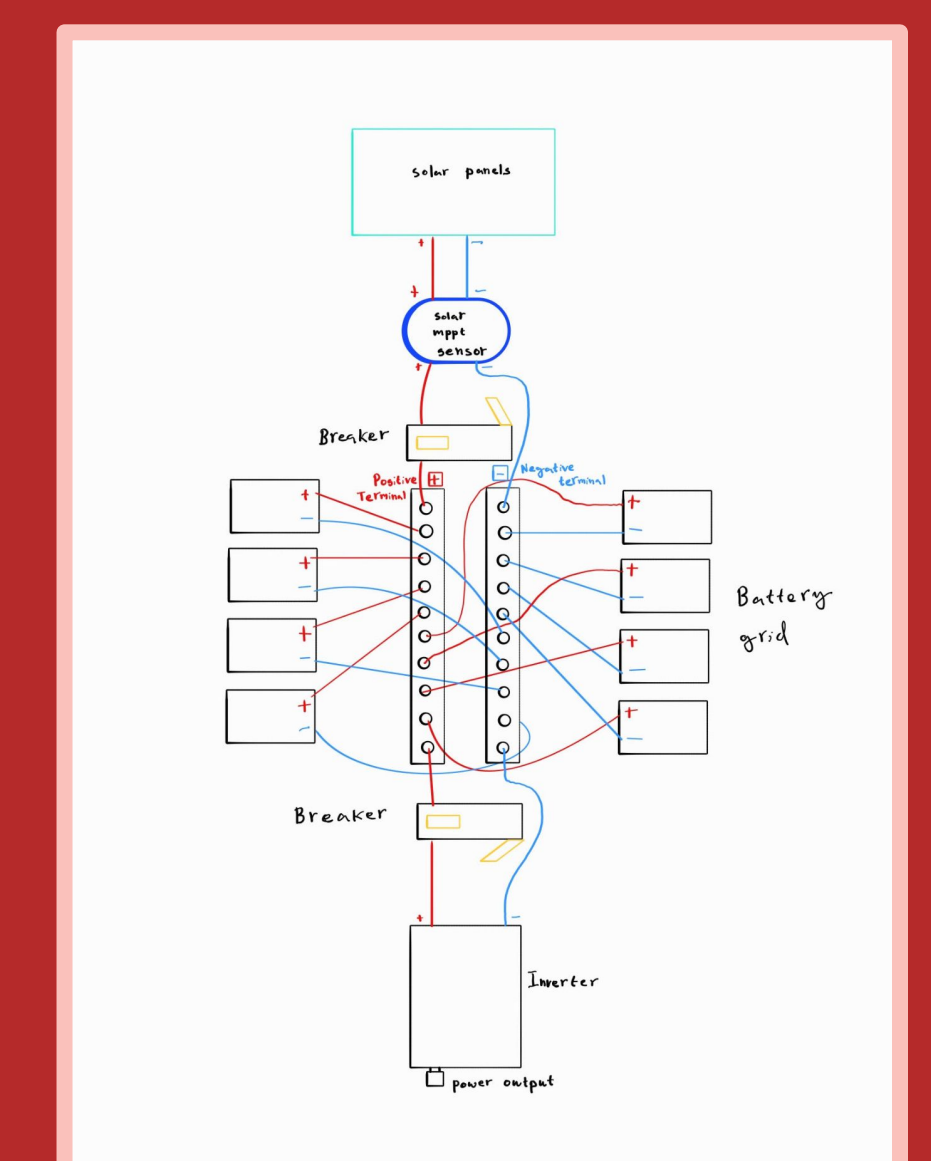


A grid-tied system is the most straightforward and cost-effective configuration.

- Solar panels convert sunlight into electricity
- Excess energy is stored in batteries and after that is sent to the utility grid
- Excess energy is turned into credits through a buyback system



Solar panels generating electricity under direct sunlight



Battery system schematic illustrating energy flow